Stage 3K Physics Note: Minimal C++ Implementation Plan

GL-with-AI project

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1 Scope and Status

Stage 3K is a planning stage. It does not implement C++, finite differences, source terms, or production evolutions. Future implementation should live in repo-owned code, likely under `code/BlackStringToy/` or a new repo-owned layer copied into scratch GRChombo trees under ignored `runs/`. It must not modify `external/GRChombo`.

Compilation is not evidence of physical correctness. Algebra-helper fixtures passing is not integration correctness, and green helper tests do not prove that variables are read from the correct grid slots. No Stage 4A result should be described as "cartoon evolution works" or "CCZ4 works." Every implementation stage must have a validation gate.

Stage 4A C++ work requires explicit user approval after Stage 3K review. Stage 4B and later stages also require explicit approval before source edits.

2 Convention Policy

The implementation convention target is GRChombo compatibility plus internal consistency. External implementations may be used later for validation, but they are not the source of truth for variable naming, evolved-variable basis, trace convention, or cartoon-extension bookkeeping. When public GRChombo does not define the reduced-cartoon extension directly, the project must document the extension explicitly and use it consistently.

3 Recommended First Stage

The recommended first C++ implementation stage is Stage 4A: a repo-owned conformal-cartoon algebra helper layer plus local, non-grid fixture tests. It must not connect that helper to grid variables or time evolution. This is the least risky first stage because it tests the bookkeeping that all later source terms depend on while leaving the public GRChombo RHS untouched.

Stage 4A should cover:

- the off-diagonal determinant
$$(h_{xx}h_{zz} - h_{xz}^2)h_{ww}^2;$$
- the reduced inverse conformal metric;
- full 4D trace and trace-free projection with hidden multiplicity $N_{\text{hidden}} = 2$;
- the GR_SPACEDIM $= 4$ trace denominator $1/4$;

- reconstruction of K_{ij}, including

$$K_{xz} = \chi^{-1} \left(A_{xz} + \frac{1}{4} h_{xz} K \right);$$

- the diagonal limit back to the Stage 3F formulas.

Stage 4A explicitly excludes cartoon Ricci helpers, small-axis helpers, grid-variable wiring, CCZ4 RHS terms, gauge/Gamma-driver terms, constraint damping, finite-difference stencils, and evolution. Passing Stage 4A only shows that local algebra formulas were implemented correctly on supplied local values.

The Stage 4A review follow-up keeps that scope unchanged while making the local fixture more robust: the conformal-factor power should be expressed as $-1/\mathrm{GR\_SPACEDIM} = -1/4$ using the physical spatial dimension, local floating-point zero guards should use tolerances, and at least one independently derived K_{ij} reconstruction oracle should be included.

4 Stage 4B Public Layout Baseline

Before editing variable enums or wiring helpers to grid variables, perform a read-only baseline check of the public CCZ4 variables retained for comparison. Stage 4B proves that this public GRChombo-facing layout has not drifted unexpectedly. It does not by itself prove that h_{ww} or A_{ww} are correctly placed.

Stage 4C adds the real hidden enum names and header-level checks in `UserVariables.hpp`. With the Stage 4C names c_{hww} and c_{Aww}, the assertions are

$$c_{\mathrm{hww}} = c_K - 1, \qquad c_{\mathrm{Aww}} = c_\Theta - 1,$$

using compile-time assertions in the variable header. Stage 4D remains the next narrow step, and only after Stage 4C passes review and gets explicit approval: give h_{ww} and A_{ww} finite scaffold values so the cheap smoke run no longer dies immediately from NaNs. This is not a physical evolution check. A broader helper-to-grid handoff remains a later stage.

5 Deferred Blocks

Stage 4A should not implement the full CCZ4 RHS, gauge or Gamma-driver equations, constraint damping, black-string initial data, turduckening, finite-difference axis treatment, apparent-horizon finding, waveform extraction, production runs, or comparison with external/reference implementations.

6 Validation Map

Stage	Required validation	Source stage
Stage 4A determinant/inverse	normalized determinant, block inverse	3F–3G
Stage 4A trace projection	hidden multiplicity and 1/4 guard	3F–3J
Stage 4A K_{ij} reconstruction	$K_{xx}, K_{xz}, K_{zz}, K_{ww}$ round trip	3F–3G
Stage 4B public baseline layout	public CCZ4 visible enum/name order	3J–3K
Stage 4C hidden enum placement	header-level h_{ww}/A_{ww} assertions	3J–3K
Stage 4D finite scaffold values	h_{ww}/A_{ww} no longer fail cheap smoke with NaNs	3J–3K
Later grid handoff	helper input slots receive intended live components	3J–3K
Stage 4E cartoon Ricci helper	flat, round S^2, warped, sheared-flat gates	3C–3G, 3J
Stage 4F small-axis helper	assembled $\hat{\Gamma}^x$ limit guard	3I–3J
Later constraint damping	injected-constraint linearized milestone	3H–3J

7 Primary Risks

The first implementation must guard against using a 2D trace instead of a full 4D trace, omitting h_{ww} from determinant or trace operations, double-counting the external x^2 factor by treating h_{ww} as $g_{\theta\theta}$, using reciprocal inverses when $h_{xz} \neq 0$, leaving a residual $1/x$ in $\hat{\Gamma}^x$, trusting flat-zero checks for constraint damping, wiring correct helper formulas to the wrong enum slot, editing `external/GRChombo`, or trusting compilation alone.

8 Acceptance

Stage 3K is complete when the minimal first implementation stage is limited to local algebra helpers and non-grid fixtures, deferred blocks are listed, repo-owned implementation targets are mapped without editing code, each stage is linked to Stage 3J tests, automated layout gating is required before grid wiring, and no C++ implementation has started. Stage 4A is the approved follow-up stage that implements the local helper and non-grid fixture only.